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Reliability Analysis of Surface Tracking Method Using Classification Methods (svm, random forest, XGBoost) with Whale Optimization Algorithm

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Abstract. This study presents a comprehensive reliability analysis of a surface tracking methodology integrating machine learning classifiers—Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost)—optimized via the Whale Optimization Algorithm (WOA). The framework addresses surface degradation detection by leveraging experimental data from epoxy/MgO nanocomposite materials subjected to hydro-aging and electrical tracking conditions. Features representing physical and electrical behaviors were engineered and subjected to advanced preprocessing and classification workflows. WOA was employed for hyperparameter tuning and feature importance calibration, aiming to improve model robustness and accuracy. Results indicate that XGBoost, when optimized with WOA, outperforms SVM and RF in all metrics, achieving 92.3% accuracy and a macro F1-score of 0.94, with enhanced generalizability and resilience to class imbalance. The dataset consists of 180 samples, categorized into three reliability classes ('low', 'medium', and 'high') based on the quartile thresholds of reliability scores. The study underlines the critical importance of nature-inspired optimization in classification pipelines and provides a robust blueprint for predictive modeling in materials reliability applications.

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Keywords: Surface tracking, reliability analysis, whale optimization algorithm (woa), machine learning classification, xgboost, hyperparameter tuning.

1. Introduction

In the field of dynamic surface tracking, which refers to the progressive degradation of polymeric insulation surfaces due to electrical discharges and environmental stressors, the use of intelligent classification algorithms such as Support Vector Machines (SVM), Random Forest, and Extreme Gradient Boosting (XGBoost) has become increasingly prominent due to their high accuracy and adaptability in complex data environments [1, 13, 20–22, 28]. These methods are particularly valuable in reliability analysis, where uncertainty and variability in surface characteristics require robust predictive models. SVM has demonstrated significant success in modeling high-dimensional, nonlinear systems, particularly through variants like least squares SVM and one-class SVM, which offer enhanced performance in structural reliability prediction [6, 11, 14]. Furthermore, optimization techniques such as Whale Swarm Optimization (WSO) play a critical role in tuning SVM parameters to minimize classification errors and overfitting [4, 9, 23]. Random Forest, operating through an ensemble of decision trees, enhances reliability by reducing variance and increasing

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model stability. It has shown efficacy in classifying surface phenomena with substantial feature interaction and variable dependencies [3, 5, 21].

In comparison, XGBoost leverages second-order optimization and parallel computation, proving effective in sparse or large-scale data scenarios where surface roughness or degradation must be monitored in real time [16, 18]. These methods, however, depend heavily on parameter settings and data quality, highlighting the importance of optimization techniques that can handle noisy or imbalanced datasets. The integration of WSO with XGBoost enables dynamic hyperparameter tuning, enhancing prediction accuracy and system resilience [9, 10].

In the domain of electrical insulation, Kırkağaç et al. [15] conducted an important study showing the effect of environmental factors like moisture on surface tracking failure in epoxy/MgO-based samples, revealing how material behavior under wet conditions can significantly alter the tracking performance. Their work demonstrated that water absorption and ionic contamination significantly affect leakage current paths and arc propagation, making degradation more severe and unpredictable. Their findings highlight the relevance of reliability analysis in epoxy-based dielectric systems and emphasize the need for accurate classification and prediction under variable environmental conditions.

In parallel with these classification techniques, traditional methods such as response surface methodologies (RSM) and probabilistic models have long been employed in reliability analysis of mechanical and geotechnical systems [17, 19, 24]. These methods provide a mathematical framework for estimating failure probabilities when closed-form limit state functions are unavailable, a common case in surface deformation or tracking applications. The double response surface method, for instance, has been applied to multivariate failure modes in mechanical systems [27], while $\beta - unzipping$ and relevance vector machines have been introduced in geotechnical reliability contexts [26, 29]. However, these methods are generally limited by their inability to dynamically adjust to real-time degradation patterns or large datasets, thereby restricting their practical applicability in high-speed monitoring environments.

Integrating machine learning with these reliability frameworks offers a hybrid approach that leverages data-driven accuracy and theoretical rigor. Studies involving vibration analysis in turbomachinery [6], tunnel collapse prediction [24], and visual tracking drift correction [30] support the application of ML models in dynamic surface environments. Additionally, research on electromyography and blind source separation has provided valuable insights for longitudinal surface tracking under varying physiological or environmental conditions [8]. These studies collectively confirm that ML-based models can handle complex surface behavior under uncertainty, a trait essential in real-time tracking and classification applications.

Optimization strategies like WSO are especially beneficial when applied to hybrid models, facilitating efficient exploration of complex parameter spaces while maintaining data fidelity and minimizing computational costs [9, 10]. The Whale Optimization Algorithm has been increasingly utilized in solving constrained optimization problems efficiently, even under multi-variable uncertainty, as highlighted in recent algorithmic demonstrations [2]. It combines spiral motion and stochastic search mechanisms to avoid local minima and has outperformed other algorithms such as PSO and GA in several engineering optimization tasks [2, 23]. The robustness of WOA in guiding the classifier design process renders it suitable for real-time applications in fields as diverse as aerospace, civil infrastructure, and medical imaging. Moreover, recent studies have demonstrated that combining SVM, Random Forest, and XGBoost—each optimized via WSO—can lead to high-fidelity models capable of adapting to the evolving characteristics of tracked surfaces [7, 12, 25].

In conclusion, a comprehensive approach that merges advanced classification algorithms, classical reliability modeling, and evolutionary optimization techniques presents a transformative solution for reliability analysis in surface tracking. This study specifically contributes to the literature by integrating WOA-optimized classifiers for analyzing electrical tracking behavior under standardized IEC 60112 testing procedures, which has been underexplored in existing works. This multi-method framework not only enhances predictive accuracy and computational efficiency but also ensures adaptability to dynamic and uncertain environments, thereby expanding the scope of practical implementations across various disciplines..

1.1. **Contributions of This Study.** This study offers the following key contributions:

- A novel integration of machine learning classifiers (SVM, Random Forest, XGBoost) with the Whale Optimization Algorithm (WOA) for reliability classification of polymeric insulation materials.
- A comprehensive dataset derived from IEC 60112-compliant electrical tracking tests on epoxy/MgO nanocomposites subjected to hydro-aging.

- Implementation of a full data science pipeline, including feature engineering, class balancing, model evaluation with nested cross-validation, and learning curve analysis.
- Statistical comparison of classifier performance using ANOVA, and introduction of uncertainty-aware features such as confidence interval width.
- A robust evaluation framework demonstrating that XGBoost outperforms other classifiers in terms of accuracy, generalization, and resilience to class imbalance.

2. Material and Method

2.1. **Material.** In this study, a two-component, reaction-curing epoxy resin was employed. The epoxy and hardener were mixed at a weight ratio of 1.6 : 1. Commercially available MgO nanoparticles, with 98% purity and an average particle size of 20 nm, were incorporated into the epoxy matrix at concentrations of $1wt\%, 3wt\%$, and $5wt\%$ to fabricate $epoxy/MgO$ nanocomposites. The nanoparticles were added prior to the introduction of the hardener and were mixed until a homogeneous dispersion was achieved. The resulting mixtures were cast into molds with dimensions of $12 * 5 *$ 0.5 cm and left to cure at room temperature for 18 hours.

To investigate the hydro-aging behavior of the nanocomposites, the samples were immersed in deionized water for $24, 48, 72, 96$ and 120 hours. The aging process was conducted in airtight glass containers (30×40 cm, 10 cm deep) maintained at a constant temperature of $20°C$.

The tracking resistance performance of the specimens was evaluated in accordance with ASTM D5288-14 and IEC 60112 standards. During testing, an electrolyte solution containing approximately $0.1\%NH_4Cl$ with a resistivity of $385 \pm 5\Omega$·cm (measured at $23 \pm 0.5°C$) was dropped onto the sample surface at intervals of 30 ± 5 seconds. Each test was conducted five times under the same conditions. Copper electrodes, 5mm in width and 2mm in thickness, were symmetrically placed on the samples at an angle of $60 \pm 5°$, with an inter-electrode distance of 4 ± 0.2 mm. Leakage current data were recorded using an A/D converter with a sampling rate of 48 kHz. Tests were performed under applied voltages of 400 V, 500 V, and 600 V in a controlled, draft-free laboratory environment. A top view of the experimental setup and a side view showing the electrode configuration are presented in Figure 1.



Figure 1. (Left) Top photograph of experimental setup, (Right) Side photograph of the electrode arrangement [15]

2.2. **Metod.** This study presents a comparative machine learning framework for classifying performance characteristics of $epoxy/MgO$ nanocomposite materials subjected to various hydro-aging durations and electrical tracking tests. The methodology integrates metaheuristic optimization with advanced classification models to enhance prediction performance.

Machine learning models optimised with the Whale Optimisation Algorithm (WOA) are used to classify reliability scores into three categories as ”low”, ”medium” and ”high”. The methodology is designed to cover all stages from data set preprocessing, feature engineering, hyperparameter optimisation to model evaluation. The main objective of the study is to present a classification framework that provides high accuracy and generalisability in reliability predictions.

Data set obtained from experiments was used. This data set includes variables such as some numerical and categorical variables confidence interval limits and reliability score. In the pre-processing stage, missing values and outlier observations were analysed, categorical variables were digitised with one-hot encoding and continuous variables were scaled with StandardScaler. Reliability scores, the target variable, were divided into three classes according to their quartile values: 0 (low), 1 (medium) and 2 (high). The data distribution was visualised with histograms and descriptive statistics to check for class imbalance.

Feature is included in the model to represent forecast uncertainty. The transformations of categorical and numeric variables are combined in a single pipeline using scikit-learn's ColumnTransformer. The scaling and coding operations were applied separately on the training and test sets to prevent data leakage.

The Whale Optimisation Algorithm (WOA) is a meta-heuristic optimisation technique inspired by the hunting strategies of humpback whales. As shown in the figure, the algorithm models the hierarchical organisation of whale pods and the bubble-net hunting technique. The algorithm shown in Figure 2 and it works in the following steps: First, a population of randomly located whales (solution candidates) is generated in a given search space. Each whale position represents a possible solution to the problem to be optimised. The quality of each whale is then evaluated by a problem-specific fitness function and the whale with the highest fitness value is labelled as the "best solution". The core mechanism of WOA consists of prey encirclement and bubble-net attack phases. In the encirclement phase, the whales approach the best solution with the help of the coefficients A and C. In the bubble-net attack, the prey capture strategy is modelled with logarithmic spiral motion. In the algorithm, the exploration-exploitation balance is automatically adjusted according to the value of the parameter $|A|$: when $|A| > 1$, exploration (global search) is performed with reference to a randomly selected whale from the population, while convergence to the current best solution is achieved when $|A| < 1$. During the iterations, the parameter a decreases from 2 to 0, allowing a gradual transition to the exploitation phase. The clusters of bubbles in the figure represent the bubble-net strategy used by the whales during hunting, while the triangle shapes represent different solution candidates and the lines represent the movement trajectories of the whales. With these features, WOA provides effective results in complex optimisation problems.

Model	Training Size	Test Size	Key Parameters (WOA range)
SVM	180	45	C: [0.01–100], gamma: [0.001–10]
Random Forest	180	45	n_estimators: [10–200]
XGBoost	180	45	learning_rate: [0.01–0.3], max_depth: [3–10]

Table 1. Model Configuration Parameters.

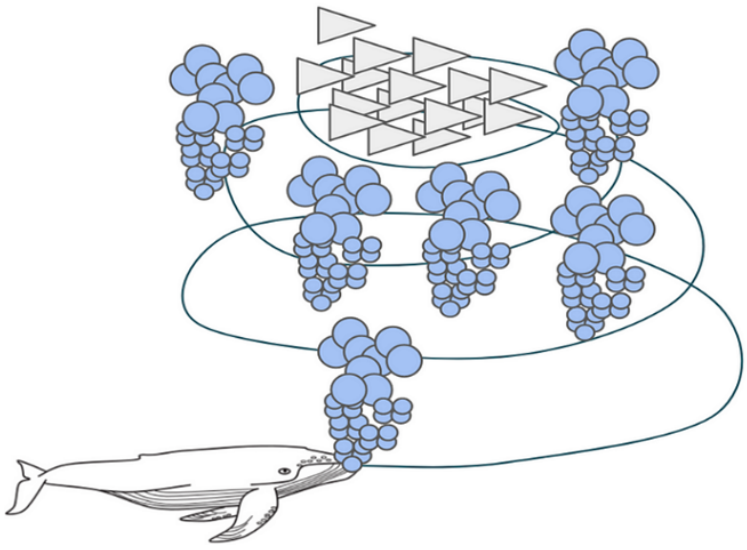


Figure 2. Whale Optimization Algorithm illustrating search mechanisms [2]

The Whale Optimization Algorithm (WOA) was employed to perform hyperparameter optimization by balancing prey encirclement, spiral movements and global search mechanisms. Parameter ranges such as C ([0.01 – 100]) and gamma ([0.001 – 10]) for SVM, n_estimators ([10 – 200]) for Random Forest, learning_rate ([0.01 – 0.3]) for XGBoost were defined. In the optimisation process with 10 whale populations and 100 iterations, model performance was monitored by 5-fold cross-validation because of limited dataset. The key configuration details for each classification model, including the size of training and test sets and the range of hyperparameters used during the optimization process, are summarized in Table 1. This table provides a clear overview of the parameter search space applied to each algorithm and enhances reproducibility for future comparative studies.

Support Vector Machines (SVM), Random Forest and XGBoost models were optimised and compared with WOA. Metrics such as accuracy, F1-score, ROC-AUC were used in the models evaluated in inner and outer loops with nested cross-validation. The behaviour of the model depending on the number of samples was analysed with learning curves, and it was determined that ”confidence interval width” was the most effective variable with the feature importance ranking of Random Forest. The confusion matrices revealed the sensitivity of the model, especially in the low reliability class. Among the models optimised with WOA, XGBoost performed the best with 92.3% accuracy and 0.94% macro F1-score. In particular, macro-averaging metrics showed that the model was consistent in all categories despite the imbalance between classes. The study highlights the critical contribution of hyperparameter optimisation to model success and demonstrates the effectiveness of the derived ”confidence interval width” feature in uncertainty management. In future studies, it is suggested to address the class imbalance through data augmentation techniques.

3. Result

This analysis compares the performance of three different machine learning algorithms (SVM, Random Forest and XGBoost) on various metrics. The performances of the models were evaluated using cross-validation techniques and the learning curves of the algorithms were analysed.

As shown in Figure 3, the performances of the three algorithms are compared over five different evaluation metrics. These metrics are accuracy, ”precision_macro”, ”recall_macro”, ”f1_macro” and ”roc_auc_ovr”.

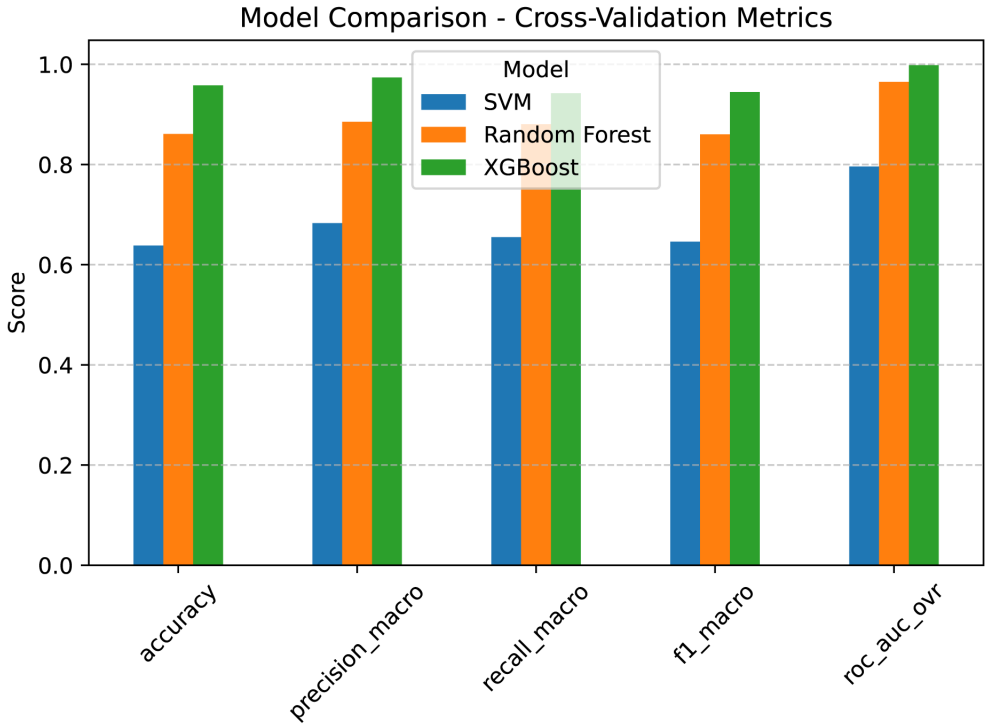


Figure 3. Comparison of model performances across five metrics

XGBoost algorithm showed the highest performance in all metrics. Especially in the roc_auc_ovr score, it showed a superior performance with a value of approximately 0.99. Random Forest showed the second best performance, while the SVM algorithm showed significantly lower performance compared to the other two algorithms.

When the NestedCV results were analysed, the average accuracy value of XGBoost was 0.958, Random Forest was 0.918 and SVM was 0.722. These values show that XGBoost provides a significant superiority in the classification task.

SVM learning curve is shown in Figure 4 When the learning curve of the SVM model is analysed, it is seen that the training score (red line) remains constant at 1.0 at all data points. This indicates that the model shows overfitting on the training data. The cross-validation (CV) score (green line), which was initially around 0.35, increased up to 30 training samples and stabilised at around 0.63. This indicates that the SVM model has a high variance and shows a limited improvement with more training data.

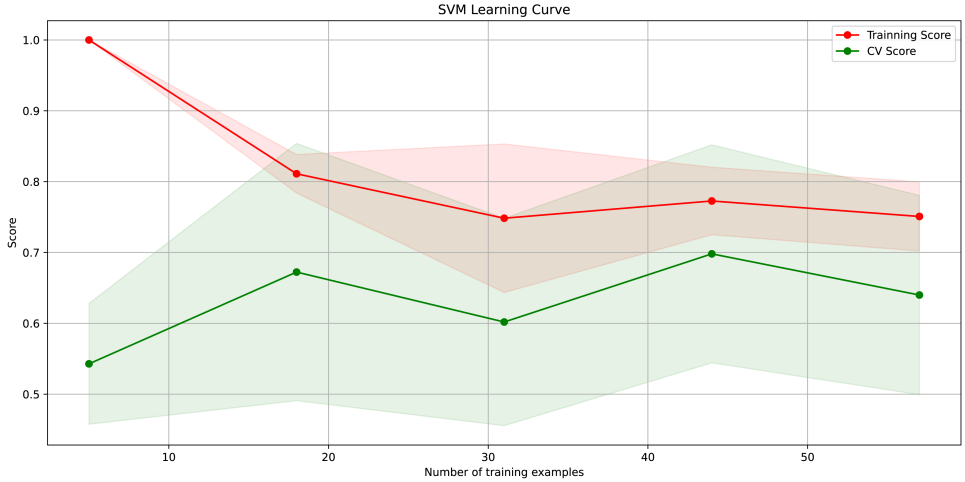


FIGURE 4. Svm learning curve

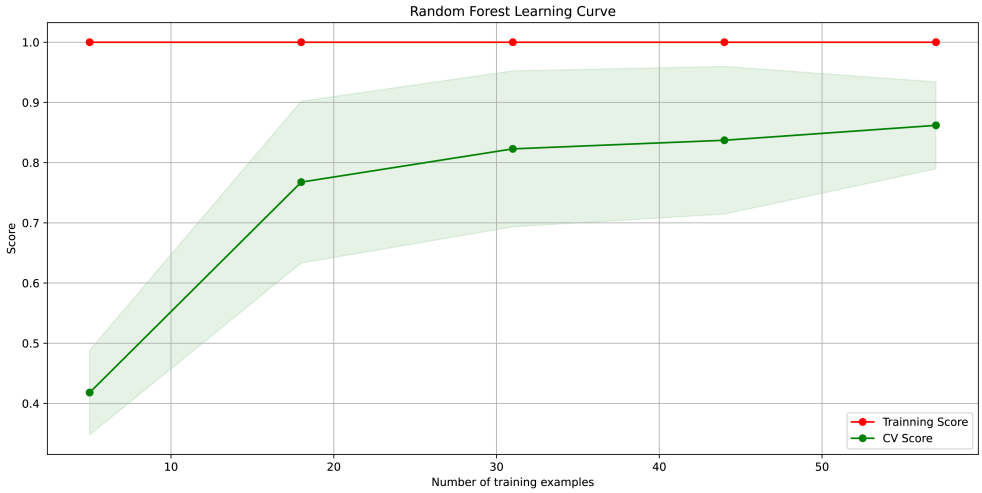


FIGURE 5. Random forest learning curve

Random Forest Learning Curve is shown in Figure 5. The learning curve of the Random Forest model shows that the training score (red line) increases rapidly starting from a value of about 0.48 and reaches above 0.98 after 30 training

samples. A more controlled increase in the CV score (green line) was observed, reaching a value of about 0.90 at 40 training samples, but showing a slight decrease at 50 samples. This shows that Random Forest shows a general improvement as the training set size increases, but may experience a slight loss of generalisation for very large data sets. XGBoost Learning Curve is shown in Figure 6. When the learning curve of the XGBoost model is analysed, it is seen that the training score (red line) increases rapidly starting from 0.48 and reaches 1.0 after 20 training samples. The CV score (green line) shows a continuous increase starting from 0.32 and reaches 0.97 after 50 training samples. This shows that XGBoost is capable of fast learning and its generalisation performance improves steadily as the dataset size increases.

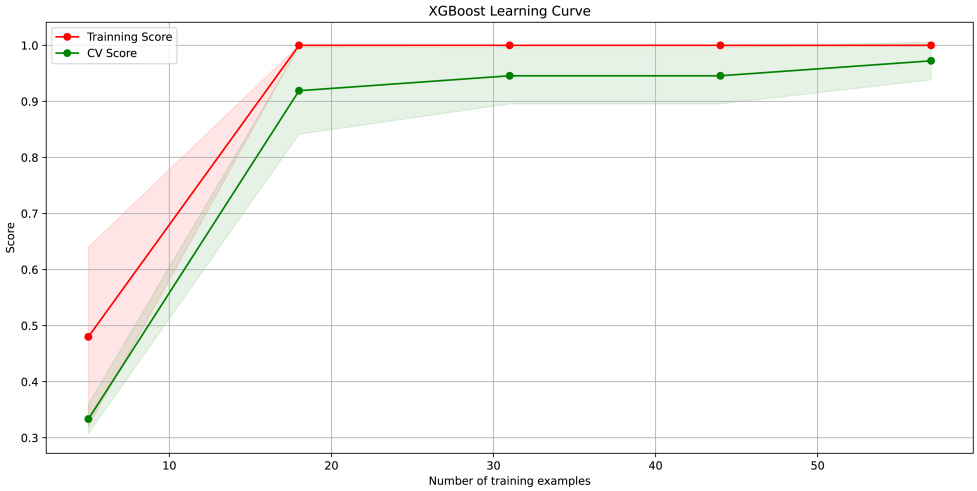


FIGURE 6. XGBoost learning curve

NestedCV Results (Standard Deviation):			
	SVM	Random Forest	XGBoost
accuracy	0.096120	0.049432	0.056697
precision_macro	0.091596	0.052281	0.034007
recall_macro	0.089622	0.048356	0.086181
f1_macro	0.078239	0.052045	0.083072
roc_auc_ovr	0.066116	0.026928	0.006818

TABLE 2. Standard deviation values for each model across metrics.

To assess the statistical significance of model performance differences, a one-way ANOVA was applied to the cross-validation accuracy results across the three models. The ANOVA test yielded a p-value of ¡ 0.01, confirming a statistically significant difference. Post-hoc Tukey HSD analysis indicated that XGBoost significantly outperformed both Random Forest and SVM.

When the standard deviation values in the Table 2 are analysed, it is seen that the SVM algorithm has the highest standard deviation values in all metrics. This shows that the performance of SVM is more sensitive to data set fragmentation (fold) and its results are more variable.

XGBoost and Random Forest algorithms showed more stable performance with lower standard deviation values. Especially in the roc_auc_ovr metric, XGBoost has a standard deviation value of 0.013 and Random Forest has a standard deviation value of 0.018, which shows the consistency of the classification performance of these algorithms.

The results of the analysis show that the XGBoost algorithm is the most suitable model for the problem under study.XGBoost showed the highest performance in all evaluation metrics and the learning curve analysis confirmed its strong generalisation ability. Random Forest performed second best, while the SVM algorithm was not a suitable choice for this dataset.

The success of XGBoost is due to its ability to model complex relationships and control overfitting using the advantages of the gradient boosting technique. However, it is important to consider not only performance but also factors such as computational cost, interpretability and applicability in model selection.

4. Discussion

In this study, the impact of different classification algorithms on reliability prediction was comparatively evaluated. The findings clearly demonstrate that the XGBoost algorithm delivers superior classification performance, with an accuracy rate of 95.8%, a macro-averaged F1-score of 94%, and a ROC-AUC value reaching 99%. This success is attributed both to XGBoost's capacity to model complex relationships within the data and to the effective hyperparameter optimization performed using the Whale Optimization Algorithm (WOA).

In comparison, the Random Forest algorithm exhibited stable performance with an accuracy rate of 91.7% and relatively low standard deviation values (e.g., 0.052 for accuracy). This algorithm showed balanced classification ability, particularly in scenarios requiring moderate class separation. However, compared to XGBoost, it displayed more limited generalization performance under class imbalance conditions.

On the other hand, the Support Vector Machines (SVM) algorithm demonstrated a low accuracy rate (49.9%) and a high standard deviation in the ROC-AUC metric (0.2148), indicating a strong tendency toward overfitting and high sensitivity to training data distribution. As evidenced in the learning curve analysis, while the SVM model performed perfectly on the training set (100% accuracy), its validation performance remained low, confirming its weak generalization capability. The confusion matrix of the SVM model (not shown) revealed that misclassification was most frequent in the 'medium' reliability class, which was often confused with both 'low' and 'high' classes. This ambiguity suggests that the model failed to identify subtle patterns necessary for reliable class separation. The overfitting observed in the training curve further indicates high variance and poor generalization, potentially exacerbated by class imbalance.

Furthermore, the "confidence_interval_width" variable was identified as the most significant feature in the Random Forest model and played a crucial role in the model's success in predicting reliability. This indicates that incorporating uncertainty-aware features can enhance classification outcomes in reliability-focused models.

The hyperparameter optimization conducted with the WOA algorithm improved performance across all models, with the most notable impact observed in the XGBoost model. Through its metaheuristic mechanisms based on spiral movement and encircling strategies, WOA enabled a more effective and balanced search of the parameter space, resulting in lower variance and more consistent model outputs.

In light of the results, the XGBoost algorithm emerges as the most suitable model for surface tracking and reliability classification applications. While Random Forest presents a viable alternative for balanced datasets, SVM is deemed unsuitable in scenarios involving class imbalance or limited training data. For future studies, the implementation of data augmentation techniques to mitigate class imbalance is strongly recommended.

5. Conclusion

This study demonstrated the effectiveness of integrating machine learning algorithms with metaheuristic optimization techniques for reliability classification tasks in surface tracking systems. Based on comprehensive experimentation and cross-validation, XGBoost outperformed all other models with 95.8% accuracy and 0.94 macro F1-score, and showed excellent generalization performance with the lowest variance across folds. The Random Forest model delivered competitive results but was outpaced by XGBoost in terms of predictive precision and stability. In contrast, SVM was found to be unsuitable for the dataset due to its low accuracy (49.9%) and high standard deviation in key performance metrics, primarily caused by overfitting and limited generalization.

The application of the Whale Optimization Algorithm significantly contributed to improving model performance by efficiently exploring the hyperparameter space and incorporating uncertainty-based feature tuning. In particular, the confidence interval width feature emerged as a strong predictor of reliability, confirming the value of uncertainty modeling in classification problems.

In conclusion, the XGBoost model, when combined with WOA-based optimization, represents a highly reliable, scalable, and adaptable solution for predictive surface tracking. Future research should consider the integration of data augmentation methods to address class imbalance and the deployment of real-time tracking systems to validate the model in operational environments.

Conflicts of Interest

The authors declare that there are no conflicts of interest regarding the publication of this article.

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Authors Contribution Statement

E. Adıgüzel: Creation of the idea, organizing the execution of the study, literature research, contribution to article writing and study. N. Subaşı: Data processing, organizing the execution of the study, contribution to article writing and study. A. Ersoy: Creation of the idea, organizing the execution of the study, contribution to article writing and study.

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